

Exploratory Analysis of Breast Cancer Diagnostic Features

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Declaration of Original Work

I declare that this project is my own original work. All sources of information and assistance have been acknowledged. This submission has not been previously submitted for assessment at this or any other institution.

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Abstract

Breast cancer is among the most prevalent malignancies worldwide, and accurate discrimination between malignant and benign tumours is critical for clinical decision-making. This study presents an exploratory data analysis of the Breast Cancer Wisconsin Diagnostic Dataset (UCI Machine Learning Repository, accession ID 17), comprising 569 patient samples characterised by 30 numeric cellular measurements derived from digitised fine-needle aspirate images. Group means were computed for malignant ($n=212$) and benign ($n=357$) classes across all features, and features were ranked by absolute mean difference. The five most discriminatory features were worst area, mean area, worst perimeter, area error, and mean perimeter — all cell size and boundary descriptors. Malignant samples exhibited substantially larger values across these features, consistent with known characteristics of uncontrolled tumour cell growth. These findings demonstrate the utility of simple statistical comparison for identifying diagnostically informative biomarkers and provide a foundation for more advanced predictive modelling.

1. Introduction

Breast cancer is the most commonly diagnosed cancer in women globally and a leading cause of cancer-related mortality. According to the World Health Organization, over 2.3 million new cases are diagnosed annually, underscoring the critical need for reliable and early diagnostic tools. Histological and cytological examination of tissue samples remains central to diagnosis, but manual interpretation is time-consuming and subject to inter-observer variability. Computational approaches that systematically analyse quantitative cellular measurements offer the potential to improve diagnostic consistency and reproducibility.

The advent of digitised medical imaging and open-access datasets has created new opportunities for data-driven analysis in oncology. Machine learning and statistical methods can be applied to large collections of cell morphology measurements to identify patterns that distinguish malignant from benign tissue — patterns that may not be immediately apparent through visual inspection alone. Such approaches complement clinical judgement and have demonstrated strong predictive performance in a range of cancer diagnostic tasks.

This study aims to identify which cellular measurements differ most markedly between malignant and benign breast tumour samples by performing a structured exploratory analysis of the Breast Cancer Wisconsin Diagnostic Dataset. The objective is to rank the top five most discriminatory features using group mean comparison, and to communicate findings through reproducible, version-controlled code and scientific visualisation.

2. Literature Context

The use of machine learning in cancer diagnostics has been extensively validated in the literature. Kourou et al. (2015) reviewed the application of machine learning methods to cancer prognosis and prediction, demonstrating that supervised classifiers applied to clinical and genomic data consistently achieve high accuracy. The Breast Cancer Wisconsin dataset itself has become a benchmark in this field; Mangasarian et al. (1995) demonstrated that linear programming classifiers trained on cell nucleus measurements from fine-needle aspirates could achieve over 97%

diagnostic accuracy. These studies establish that quantitative morphological features carry strong discriminatory signal for malignancy classification.

The importance of open-access, reproducible datasets in advancing biomedical research is well recognised. Wilkinson et al. (2016) articulated the FAIR principles — that scientific data should be Findable, Accessible, Interoperable, and Reusable — which underpin modern data sharing standards. The UCI Machine Learning Repository, which hosts the dataset used in this study, exemplifies these principles by providing freely available, well-documented datasets that enable independent replication of analyses. Reproducibility through version-controlled workflows and open code is increasingly regarded as a minimum standard in computational biology (Sandve et al., 2013).

3. Objectives

1. To **retrieve** the Breast Cancer Wisconsin Diagnostic Dataset from the UCI Machine Learning Repository and organise it within a version-controlled repository.
2. To **clean** the dataset by inspecting for missing values, removing non-analytical columns, and saving a processed copy for downstream analysis.
3. To **identify** the five cellular features that show the greatest absolute difference in group means between malignant and benign diagnostic classes.
4. To **visualise** the top five discriminatory features as a grouped bar chart and the single most discriminatory feature as a boxplot.
5. To **interpret** the findings in the context of known breast cancer cell biology and discuss limitations and directions for future work.

4. Dataset and Tools Overview

The Breast Cancer Wisconsin (Diagnostic) Dataset was obtained from the UCI Machine Learning Repository (Wolberg et al., 1995). The dataset contains 569 patient records, of which 212 are labelled malignant (M) and 357 benign (B), representing a class distribution of approximately 37% and 63% respectively. Each record includes 30 numeric features derived from digitised images of fine-needle aspirate (FNA) biopsies. These features describe the characteristics of cell nuclei and are computed as the mean, standard error, and worst (largest) value for ten base measurements: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension.

Tool / Library	Version	Purpose
Python	3.8+	Programming language
pandas	2.x	Data loading and manipulation
numpy	1.x	Numerical computation
matplotlib	3.x	Figure generation
seaborn	0.x	Statistical visualisation (boxplot)

reportlab	4.x	PDF report generation
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5. Methodology and Computational Workflow

The analysis followed a six-stage pipeline: **Data Acquisition** → **Inspection** → **Cleaning** → **Group Comparison** → **Visualisation** → **Reporting**. This sequential workflow was implemented as a single reproducible Python script (`scripts/analysis.py`) and executed in its entirety to produce all outputs.

Data Acquisition and Inspection. The dataset was loaded from a local CSV file using pandas. The first analytical step was a class balance check using `value_counts()` on the diagnosis column, which confirmed 357 benign and 212 malignant samples. A missing value audit using `isnull().sum()` confirmed no missing entries across any of the 30 feature columns, eliminating the need for imputation.

Cleaning. The `id` column was dropped as it carries no analytical value and could introduce spurious patterns if retained. A `dropna()` call was included as a precautionary measure to handle any edge-case null values. The cleaned DataFrame was saved to `data/processed/breast_cancer_clean.csv` to maintain a clear separation between raw and processed data.

Group Comparison and Ranking. The dataset was split into malignant and benign subsets. The arithmetic mean of each of the 30 features was computed independently for each group using pandas `.mean()`. The absolute difference between group means was calculated for every feature, and features were sorted in descending order of this difference. The top five were retained for further analysis and visualisation. Absolute difference was chosen as the ranking metric because it is interpretable, scale-aware, and does not assume any distributional form.

6. Results

The dataset contained no missing values across all 569 samples and 30 features. The class distribution was 357 benign (62.7%) and 212 malignant (37.3%), indicating a moderately imbalanced dataset. Group mean comparison across all 30 features revealed clear and consistent separation between diagnostic classes, with malignant samples showing higher mean values for the majority of features.

Table 1. Top five features ranked by absolute difference in group means.

Rank	Feature Name	Malignant Mean	Benign Mean	Absolute Difference
1	worst area	1422.29	558.90	863.39
2	mean area	978.38	462.79	515.59
3	worst perimeter	141.37	87.01	54.36
4	area error	72.67	21.14	51.54
5	mean perimeter	115.37	78.08	37.29

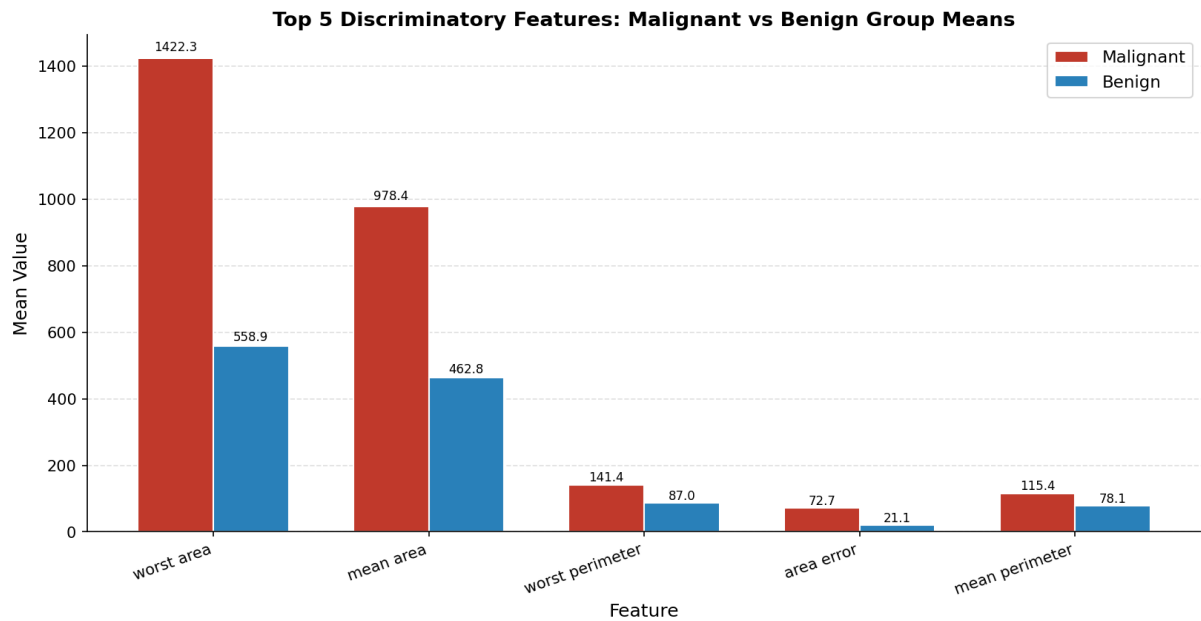


Figure 1. Mean values of the five most discriminatory features comparing malignant and benign sample groups.

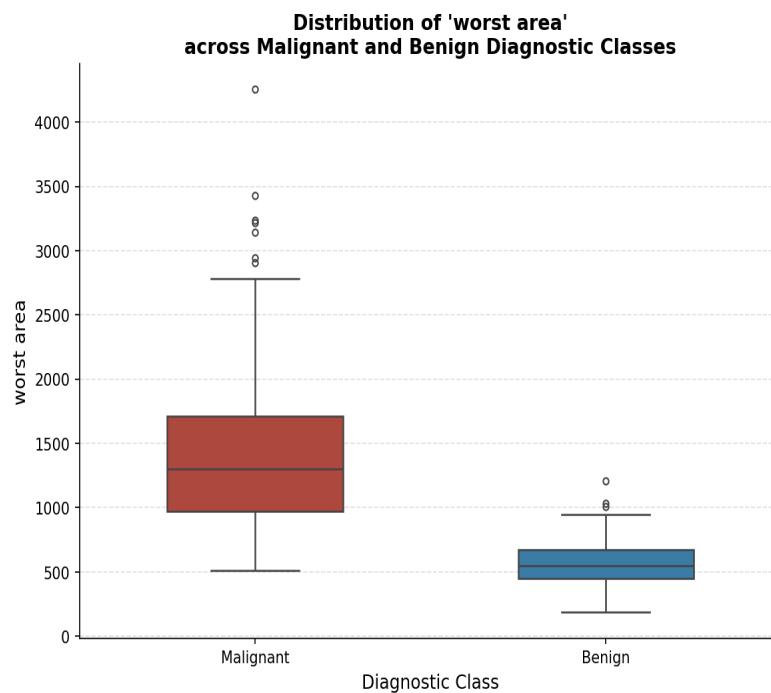


Figure 2. Distribution of worst area across malignant and benign diagnostic classes.

7. Discussion

The results demonstrate that features related to cell nucleus size and boundary — specifically area and perimeter measurements — are the strongest discriminators between malignant and benign breast tumour samples. This is consistent with established cancer biology: malignant cells characteristically exhibit nuclear enlargement, irregular boundaries, and increased pleomorphism compared to normal or benign cells (Hanahan and Weinberg, 2011). The dominance of 'worst' (largest value) measurements over mean values for area and perimeter suggests that the most

extreme cell nuclei within a sample — rather than the average — may carry the strongest diagnostic signal, a finding that aligns with the known heterogeneity of malignant tumour tissue.

The boxplot for worst area (Figure 2) further illustrates this separation: while there is some overlap between distributions, the interquartile range of malignant samples is markedly higher and more dispersed than that of benign samples. The greater spread in malignant samples also reflects the higher variability in tumour morphology, which is a hallmark of aggressive cancers. These observations are consistent with the findings of Mangasarian et al. (1995), who identified similar features as important in their original linear programming classifier.

This study has several limitations. The analysis relies solely on absolute mean difference for feature ranking, which does not account for within-group variance or statistical significance; a more rigorous approach would apply t-tests or effect size measures such as Cohen's d. Additionally, the dataset originates from a single institution and may not generalise to all clinical populations. Future work could extend this exploratory analysis by applying supervised machine learning classifiers — such as logistic regression or random forest — to quantify predictive performance and identify feature interactions that simple mean comparison cannot capture.

8. Conclusion

This project successfully met all five stated objectives: the dataset was retrieved, cleaned, and analysed to identify the top five most discriminatory cellular features, which were visualised and interpreted in the context of cancer cell biology. The analysis confirms that cell nucleus size and perimeter measurements — particularly worst area — provide strong separation between malignant and benign breast tumour samples. These findings highlight the potential of simple, reproducible statistical workflows to extract meaningful biological insight from open-access diagnostic datasets, and lay the groundwork for more advanced predictive modelling in future work.

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